

# **Absolute Space and Forward-Only Time:**

## *The Goal of This Project, and What Has Been Verified Toward It*

Daniel M. Andreola, Jr.

Independent Researcher

Date of Record: July 28, 2026

### **Status of This Document**

This document states the goal of this project plainly, in one place, and summarizes what has and has not been independently verified toward it across the companion technical documents. It introduces no new derivation of its own; every claim below is supported by a specific citation to a result already derived and checked in a companion document. Where a claim remains open or unverified, that is stated directly rather than implied by omission.

### **The Goal, Stated Plainly**

The goal of this project is narrower than building a new predictive theory of physics. It is to establish that two specific classical commitments -- that space is absolute (a real, fixed, observer-independent structure, not a convention) and does not curve or warp near mass, and that time is not a fourth dimension but a forward-only ordering of events -- are sufficient, without exception found to date, to reproduce every real, independently measured prediction of standard relativity so far tested against them. No additional spatial dimension, no curved four-dimensional manifold, and no unexplained constant introduced solely to force a match, has been required in any result listed below.

This is a claim about which foundational assumptions are necessary to reach known, verified physics, not a claim to have discovered new physics beyond it. Where this distinction matters for a specific result, it is stated explicitly in the section below.

### **What Has Been Verified**

#### **Time Dilation and the Lorentz Factor.**

Derived directly from a fixed-rate internal process (no additional dimension), using elementary geometry alone. Verified against the operational GPS relativistic clock correction (+38.5 microseconds/day predicted vs. +38 microseconds/day applied) [1, 2], cosmic-ray muon survival to sea level [3, 4], the Ives-Stilwell transverse Doppler experiment [5, 6], and gravitational redshift via the Pound-Rebka experiment [7].

#### **Magnetism.**

Derived as a relativistic consequence of electric charge and motion in three-dimensional absolute space, with no independent magnetic field posited. Reproduces the standard magnetic force law and field formula for a current-carrying wire [8].

#### **Planetary Orbital Precession.**

The flow speed governing this and the following two results,  $v_{\text{flow}}(r) = \sqrt{2GM/r}$ , is independently derived from this project's own postulate of eternal, undirected motion, combined with ordinary energy conservation -- not adopted from prior literature. Applying ordinary special-relativistic length contraction to this flow speed -- the same physical reasoning underlying Einstein's own rotating-disk argument, applied here to this derived flow speed rather than to a spinning disk -- gives a spatial term shown to be algebraically identical to the spatial component of the Schwarzschild metric, without requiring that space itself be curved.

Verified against the real measured perihelion precession of Mercury (42.99 vs. 42.98 arcseconds/century), Venus (8.63 vs. 8.62), and Earth (3.84 vs. 3.84) [9, 10].

### **Light Bending.**

The same derived spatial term, applied to a photon's relativistic momentum-energy relation, reproduces the full gravitational deflection of starlight grazing the sun (1.7500 arcseconds), matching the real value confirmed by the 1919 Eddington eclipse expedition and by modern radio interferometry [10, 11, 12].

### **The Shapiro Time Delay.**

The identical mechanism reproduces the real time delay of a radar signal passing near the sun (approximately 247 microseconds for an Earth-Mars round trip), matching the value confirmed by the Viking Mars lander and the Cassini spacecraft [10, 13, 14, 15].

### **The Black Hole Event Horizon and Tidal Field.**

The radius at which the derived flow speed reaches its fixed maximum rate is numerically identical to the standard Schwarzschild radius; differentiating the flow speed twice reproduces the standard tidal-force formula exactly [16].

### **The Innermost Stable Circular Orbit and the Photon Ring.**

The non-spinning innermost stable orbit, and the exact photon-sphere relation, are independently derived from the same flow speed and checked directly against the Event Horizon Telescope's 2019 photograph of the black hole in M87, within stated observational uncertainty [16, 17].

## **What Remains Open**

This project does not currently offer a prediction that diverges numerically from standard relativity; every result above matches standard theory rather than exceeding it [9]. Several results rely on established, independently-verified tools of physics adopted rather than derived from this project's own postulates -- the equivalence principle, and the geodesic (extremal-time) principle used to combine derived time and space terms into a single orbital equation -- and this is stated explicitly at each point of use in the companion documents rather than left implicit.

The Hulse-Taylor binary pulsar's orbital decay rate from gravitational-wave emission -- a dynamical test, historically the one that separated genuine alternatives to general relativity from those that ultimately failed -- has since been attempted, and is recorded in the companion Gravitational Radiation Investigation. That investigation resolved the required tensor shape of the radiation as a citable mathematical necessity, proved this shape requirement holds regardless of which physical force is involved, and found that this project's original four postulates do not extend to quantum mechanics: every classical combination of the quantities available (mass, the fixed rate, gravitational and electromagnetic constants) produces at most a quantity with units of energy, never the units of action on which quantum mechanics is built, confirmed as a structural limit rather than a gap awaiting the right combination. This led to the adoption of a fifth postulate -- a fixed floor on physical action, supplied by  $\hbar$ , with the same transparent, adopted status as  $G$  -- required to reach the radiation's magnitude, which remains an open, actively pursued question. A further standalone note, Neutron Star Radius from the Fifth Postulate, tests this same fifth postulate against a real neutron star's independently measured physical size, using only ordinary gravity and the uncertainty principle with no curved spacetime, and finds it correct to within observational uncertainty -- a fifth real-world confirmation of this project's foundation.

A single, precisely identified point --  $r = 0$  at the center of a collapsed mass -- remains an open singularity in this framework, exactly as it remains open in standard general relativity; this project's flat-space treatment does not resolve it, and does not currently claim to.

## References

- [1] Ashby, N. and Weiss, M., Global Positioning System Receivers and Relativity, NIST.
- [2] Fliegel, H. F. and DiEposti, R. S., GPS and Relativity: An Engineering Overview.
- [3] Rossi, B., and Hall, D. B. (1941). Variation of the Rate of Decay of Mesotrons with Momentum. *Physical Review*, 59, 223-228.
- [4] Frisch, D. H., and Smith, J. H. (1963). Measurement of the Relativistic Time Dilation Using  $\mu$ -Mesons. *American Journal of Physics*, 31(5), 342-355.
- [5] Ives, H. E., and Stilwell, G. R. (1938). An Experimental Study of the Rate of a Moving Atomic Clock. *Journal of the Optical Society of America*, 28(7), 215-226.
- [6] Ives, H. E., and Stilwell, G. R. (1941). An Experimental Study of the Rate of a Moving Atomic Clock, II. *Journal of the Optical Society of America*, 31(5), 369-374.
- [7] Pound, R. V., and Rebka, G. A. (1959). Gravitational Red-Shift in Nuclear Resonance. *Physical Review Letters*, 3(9), 439-441.
- [8] Purcell, E. M., *Electricity and Magnetism* (Berkeley Physics Course, Vol. 2).
- [9] Andreola, D. M., Jr. (2026). An Absolute-Space Derivation of Relativistic Time Dilation and the Relativistic Origin of Magnetism (revised).
- [10] Hamilton, A. J. S., and Lisle, J. P. (2008). The River Model of Black Holes. *American Journal of Physics*, 76(6), 519-532.
- [11] Andreola, D. M., Jr. (2026). Empirical Verification Ledger (revised).
- [12] Dyson, F. W., Eddington, A. S., and Davidson, C. (1920). A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919. *Philosophical Transactions of the Royal Society A*, 220, 291-333.
- [13] Andreola, D. M., Jr. (2026). Light Bending and the Shapiro Time Delay: An Absolute-Space Derivation.
- [14] Reasenberg, R. D., et al. (1979). Viking Relativity Experiment: Verification of Signal Retardation by Solar Gravity. *Astrophysical Journal Letters*, 234, L219-L221.
- [15] Bertotti, B., Iess, L., and Tortora, P. (2003). A Test of General Relativity Using Radio Links with the Cassini Spacecraft. *Nature*, 425, 374-376.
- [16] Andreola, D. M., Jr. (2026). The Black Hole Center as Exposed Absolute Space: A Working Note.
- [17] Event Horizon Telescope Collaboration (2019). First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole. *Astrophysical Journal Letters*, 875(1), L1.